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Workpackage WPB

Implementation – Online Job Vacancies

Interim report

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Summary

This report summarises the progress made with the Big Data ESSNet WPB 2018-2020 on online job vacancies implementation work package during the first 11 months of the project (November 2018 to October 2019). This summary describes the main objectives and discusses work done to date and the issues identified. These are still a work in progress. Detailed reports on analysis of the European Centre for the Development of Vocational Training (CEDEFOP) data for the participating countries are provided in the Annex.

Introduction

The aims of implementing this work package are to produce statistical estimates on the topic of online job vacancies, to identify statistical production processes and capabilities that may be affected at the national level and to define the conceptual production processes at the national level and at the level of the ESS. Suitable techniques and specific methodologies were developed during the pilot phase of the project ESSNet Big Data 2016-2018 on Web Scraping for Job Vacancy Statistics. The implementation phase is based on work carried out to establish the conditions under which web scraping techniques can be used. And also to evaluate the quality of the scraped data as well as the use of mixed sources, e.g. job portals and job adverts on enterprise websites.

It was concluded [2] that Online Job Advertisements (OJA) data cannot be used to directly replace the existing job vacancy statistics required by the EU regulation. The quality issues are such that it is not clear if these data could be integrated in a way that would enable them to meet the standards expected of official statistics. On the other hand, OJA data can provide many insights that official estimates cannot. Statistics based on OJA data can be published in a very timely and frequent manner, allowing for short-term tracking of labour market conditions and flash-estimates of labour demand. Because of the larger data pool available, OJA data also allow for more granular analysis by subgroups or geographical regions. Additionally, OJA data might allow for the provision of completely new (to official statistics) labour market insights: for instance, indicators of labour market power by employers[1] (compare [3]). There is, however, further need to address the challenges how OJA data should be interpreted and used together with official estimates.

ESSNet Big data I on Web Scraping for Job Vacancy Statistics has established a close working relationship with CEDEFOP. CEDEFOP is developing a system for all Member States and it has been agreed that this should also aim to serve the long-term needs of the ESS. The cooperation with CEDEFOP will ensure that NSIs may be able to reduce their activities around data access and data handling and focus more on the challenges around further methodological development. Therefore, NSIs should generally avoid investing heavily in developing

[1] Additionally, OJA data might allow for the provision of completely new (to official statistics) labour market insights: for instance, indicators of labour market power by employers, which can be measured by Herfindahl Hirschman Indices (HHIs) for labour markets. Azar et al (2018) calculate such a measure of labour market concentration using HHIs for labour markets at the occupation (6-digit SOC, Standard Occupational Classification), commuting zone and quarterly level based on the market share of firm j in market m. "The market share of a firm in a given market and time is defined as the sum of vacancies posted by a given firm in a given market and time divided by total vacancies posted in that market and time." [Azar et al (2018) p. 6]

web scraping approaches as OJA data are expected to become widely available to EU Member States via CEDEFOP by the end of 2020.

Organisation of work

Twelve countries participate in the work package coordinated by Slovenia (SURS):

- Germany (DE)
- Bulgaria (BG)
- Sweden (SE)
- Poland (PL)
- Italy (IT)
- Romania (RO)
- United Kingdom (UK)
- France (DARES)
- Switzerland (CH)
- Slovenia (SI)
- Lithuania (LT)
- Ireland (IR)

The work is organised under three main tasks:

Task 1 - Methodological framework

This task will produce generalized and extended methods, procedures and implementation requirements for using OJA data in statistical production. The work will be based on results of the pilot phase of the project use-cases.

Task 2 - Statistical output

This task will identify and produce statistical indicators using OJA data.

Task 3 - Implementation requirements of prototypes in the relevant statistical production processes at European and national level

Since CEDEFOP is developing a pan-European system for collecting and processing, ESSNet collaborates with CEDEFOP on using their data with the focus on facilitating the integration of these data in the future Web Intelligence Hub (WIH). WIH is developed by Eurostat as part of the Trusted Smart Statistics initiative (TSS), with the focus on the developments of CEDEFOP on Online Job Advertisements (OJA) [5].

Collaboration with the Work Package on Enterprise Characteristics (WPC) regarding methods for web scraping, data processing and analysis and the Work Package on Architecture (WPF) on designing and adopting application and information architectures was established.

Due to the large number of partners and easier organization of work, the group has set up a wiki page and organises monthly virtual meetings. Two physical meeting have been planned. The first was executed in cooperation with CEDEFOP, Eurostat and ESSNet partners, hosted by CEDEFOP in Thessaloniki from 11 to 12 June 2019.

Cooperation with CEDEFOP

During the kick-off meeting in Vienna in December 2018 WPB participants and representatives from CEDEFOP and Eurostat participated in the discussion focused on the opportunities and interests of using the CEDEFOP data. All present WPB partners expressed interest in using CEDEFOP data. It was agreed that CEDEFOP would provide all necessary information and documentation for access to and use of their data as well as related documentation (description of available data and metadata). The Big Data Task Force proposed at its meeting in April 2019 to focus on developing experimental statistics using available CEDEFOP data and the needs of the future ESS Web Intelligence Hub.

In the following months CEDEFOP, Eurostat and the WPB project coordinator tried to find a suitable solution for enabling access to and analysis of the CEDEFOP data. After several virtual meetings between CEDEFOP, Eurostat, ESSNet and CRISP it was decided that clean (pre-processed and classified) data will be available for the ESSNet using a Big Data Test Infrastructure (BDTI). The individual data processed by the CEDEFOP pan-EU system for collecting and analysing online job vacancies from 28 EU Member States are considered a basis for the work of ESSnet Big Data II WPB on Online Job Vacancies.

Alongside with publicly accessible window to the data Skills Online Vacancy Analysis Tool for Europe (OVATE), CEDEFOP has developed a Data Lab allowing users to navigate and work with the primary processed data.

WPB physical meeting and training session on how to navigate CEDEFOP's OJA data

During the first day of the first WPB physical meeting a training session on how to navigate CEDEFOP's OJA data using Big Data Test Infrastructure was organised by Eurostat, CEDEFOP, Interuniversity Research Centre on Public Services (CRISP) and Tabulaex [4].

Tabulaex presented an overview of the methodological and technical approach used with examples and results from the early release dataset. The main aim was to present data ingestion techniques, data processing pipeline, classification techniques, and data analysis path. The data ingestion phase includes all the activities related to data collection. Crawling, fetching, scraping and storing activities are the main tasks of this phase. The data processing phase includes all the activities related to preparing data for further analysis. Data preparation, translation, data cleaning and text processing tasks are the main activities of this phase. The

information extraction phase is related to the extraction and classification of structured data (occupation, skills, location, industry, etc.) from unstructured texts. The system collects data from 530 sources with over 40 million estimated OJA (daily snapshot). The data ingestion phase obtains and imports data from web portals and stores them into a database. Direct agreements with the most relevant sources exist. The main issues related to data ingestion were presented (how to handle a huge amount of near real time data, the need to detect and reduce noise, multi-language environment, the need to relate to classification standards, robustness of the process, quality of data collected, scalability and governance) as well as their solutions (redundant sources, scalable infrastructure, custom monitoring tool developed, language dependant classification framework) to assure robustness of the process, quality of data collected and scalability and governance.

The training was continued with practical exercises where participants learned how to use dashboards for different stakeholders and how to access the Data Lab for consulting data. Representatives of Tabulaex (Data Lab administrators) presented the structure of related data sets.

In the last session, ESSNet and other participants accessed the data by using DataLab Hands on. The workshop was prepared and carried out by Tabulaex experts. Tabulaex prepared guidelines, Jupyter notebook, with examples on how to access the data and run various queries in order to understand the data.

During the training, participants gained information needed to understand and use the CEDEFOP's OJV system in order to start analysing available data to identify strengths/weaknesses of the data, potential methodological issues and additional information need from CEDEFOP. On the second day of the Thessaloniki meeting, the WPB physical meeting was organised. The aim was to present and discuss future perspectives of OJA in the context of implementing future ESS Web Intelligence Hub. The main objective is to facilitate OJA data integration into statistical production by identifying and producing statistical estimates and experimental statistics and defining conceptual OJA production processes at the national level and at the level of the ESS. The aim is therefore to develop and test the methodology and statistical software.

In the discussion the role of the CEDEFOP system was discussed. The system needs to be improved continuously and sharing knowledge and solutions seems to be the best approach. For successful integration a gradual approach is needed. The aim of this WPB is also to define potential roles of the ESS and NSIs in the process of creation, usage and maintenance of such a hub.

Since this was the first physical meeting of WPB partners, the objective was also to present each partners' work done and their future plans, discuss opportunities and constraints for successful integration of OJA data into statistical production as well as plan for related future activities and requirements for additional information from CEDEFOP.

Using OJA has a number of advantages. They can be used successfully for the needs of official statistics for completing and estimating the number and characteristics of the job vacancies. Their main advantages are their relevance, timeliness, comparatively low cost of producing them and lack of or minimal burden for the

respondents. Most OJA contain information not only for the vacant position but also for the required qualifications and skills, including the so-called soft skills, for degree and kind of the graduate education and professional experience necessary for the potential applicants for taking it. Often they also contain information on the employer (sector of the economy, size of the enterprise, kind of ownership, etc.). Estimates could be made for labour demand by economy sectors, professions, specific skills and qualifications, and regions. A comparison could also be made for which job vacancies similar education, qualification and specific skills are required. However, using big data as an alternative information source for statistical surveys is always accompanied by challenges.

- Defining the scope of the survey. The Internet ads refer only to part of the actual job vacancies and not all workplaces are announced online regardless of the kind of channels. Some positions are more likely to be advertised on the Internet and the received estimates will contain deviations connected with the professions and qualification. As a rule, workplaces are announced on the Internet to a much greater extent in the high-tech economy sectors, as well as in such requiring higher education degree and specific qualification.
- There is not a certain information source for job vacancies and sometimes they are published simultaneously on different Internet channels – on specialized websites for publishing job vacancies ads, on recruitment agencies' websites, enterprises' websites, on social media, etc. Often some websites republish ads from other ones. This raises the issue of constructing methods for avoiding the so-called duplication of one and the same ad in one and the same Internet source or in different Internet sources and solving this issue is among the main challenges of using big data for monitoring job vacancies. OJAs often contain both structured and unstructured elements. In the texts of the ads in unstructured form education, qualification and skills are usually required, which should be turned into structured elements through special procedures. This requires using of specific classification algorithms adjusted for the respective language used by the published ad. However, the data received through these algorithms often contain inaccuracies and errors.
- The job vacancy ad does not always correspond to the statistical unit job vacancy.
- In the various ads the information about the vacancies is with different extent of details. Most often, the ads of the recruitment agencies do not contain employer data. This leads to the impossibility of collecting data for the enterprise sector, as well as of connecting big data to the data from the official statistical sources.

All these challenges necessitate careful searching for ways and approaches for estimating the job vacancies via information from online ads.

Based on the presentations, a discussion of what elements would be needed in order to facilitate the integration of OJA into official statistics was organised. WPB participants and Eurostat representatives were divided into two groups: methodology and statistical products, and ESS architecture and future perspectives. The methodology and statistical products group discussed and introduced a prioritised list of future activities needed:

- Establish a list of statistical indicators produced using OJA (see Table1)
- Establish test protocols
- Distribute indicators and CEDEFOP data among the participants for testing
- Compare CEDEFOP OJA data with OJV data collected at the NSI (home data)
- Prepare reports with the following components on results of the analyses describing:
    - Strengths/weaknesses of the data
    - Methodological issues raised/to solve
    - Needs for additional information from CEDEFOP
    - Recommendations for the next step
- Produce a draft for the Methodological Framework chapter

The ESS architecture and future perspectives group designed and described common building blocks of draft future ESS OJA general architectural framework.

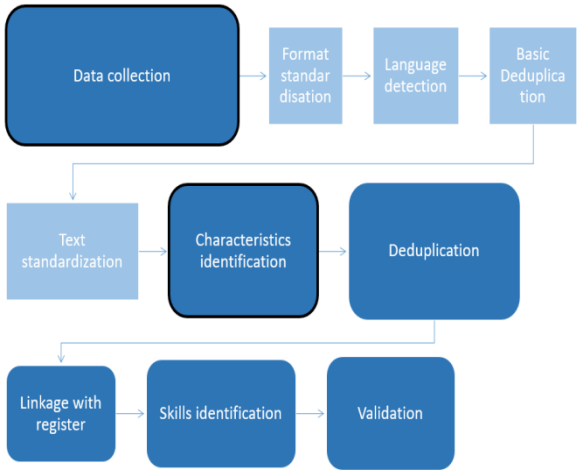

Figure 1: OJA general architecture framework

During the meeting, participants prepared a list of additional requirements towards CEDEFOP:

- Description of concepts, methods, variables, metadata of the test files
- Detailed written description of methods (ML, classifiers, deduplication rules, etc.)
- Access to the machine learning tools for testing with home data (classifiers, deduplication, etc.)
- Access to raw data before cleaning
- Access to tools/methods of collection of OJA
- Detailed description of processes for data cleaning

Use of big data test infrastructure

For the purpose of executing the training session on how to navigate CEDEFOP's OJA data and Data Lab implementation Big Data Test Infrastructure (BDTI) [6] was setup. BDTI is one of the three latest additions to the European Commission's Connecting Europe Facility (CEF) building blocks. It provides a ready-to-use cloud-based test infrastructure that enables experimentation with data analytics and the visualisation of its results. It offers public administrations in Europe the possibility to increase the adoption of big data technologies and the acquisition of analytics skills. The testing environment is complemented by a catalogue of software tools that help analyse the available datasets. During the first day of the first WPB physical meeting the DG CONNET (Communications Networks, Content and Technology) representative presented the Data Lab Architecture and available tools (Jupyter, Spark, Scikit Learn, Hue, Hive, RStudio). The practical session started with the purpose and description of BDTI infrastructure. BDTI is available as part of the CEF regulation which defines how the Commission can finance support for the establishment of trans-European networks.

Main findings of exploratory data analysis on CEDEFOP data

Between July 2019 and October 2019 CEDEFOP DataLab was available for analysis for all ESSNet WPB partners.

The focus of data analysis was primarily on the identified methodological issues to be solved and use cases already identified in the previous phases of project execution:

- Representativeness issues around online job vacancy data
- Coverage of sources and sites in data collection
- Deduplication
- Comparisons between OJAs and Job Vacancy Statistics (JVS)
- Missing information
- Selection of sources of OJA data
- Comparison of CEDEFOP and home scraped OJA (see Annex BG)

Results of the analysis show that CEDEFOP data have high potential to supplement and enrich classical statistical data, allow more granular and geo-coded insights broken down by different categories of variables (data collection period, economic sector, educational level, occupation and experience requirements). Results of the analysis demonstrate a strong relationship between OJAs and socio-economic variables (national and regional population sizes and GDPs).

Such models may potentially be used for predictive purposes.

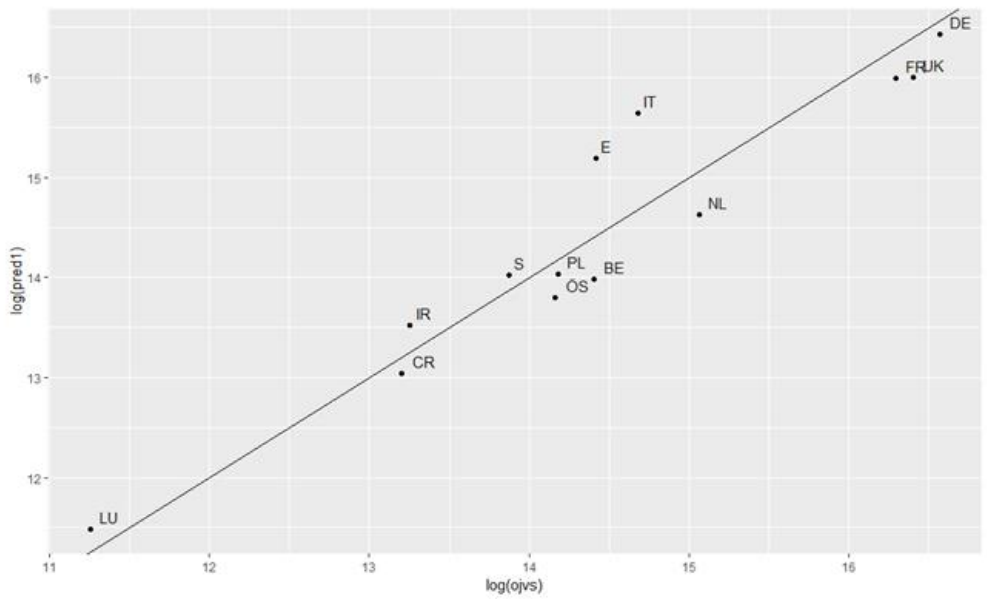

Figure 2: Scatterplot of OJVs predicted (or fitted) by the model (vertical axis) against observed OJVs for the 13 countries of the CEDFOP data (log scales)

Some consistency issues were identified between variables (sourcecountry and source), coverage of sources and sites in data collection and missing information (location, contract, source, etc.). Of particular importance for the construction of labour market indicators is the question whether CEDEFOP data represent the flow or stock of OJA. In the case of flow data, we would see a large initial stock on which to base the flows of ads. In the case of stocks, we would expect to observe similar stocks of observations for every day. Analyses indicate that it is neither. Therefore, steps have to be undertaken to calculate or approximate the stock of OJA, for instance by using pseudo-stocks (compare Annex DESTATIS). Pseudo-stocks are an approximation to the underlying stock of job ads. The idea behind pseudo-stocks is the following: At a given point in time (T), we consider a job ad to be valid if it hasn't expired yet and was posted a maximum of 30 days before T. Expiration of a job ad is determined by the variable expire_date. Preliminary analysis of other sources indicates that job ads stay relevant for roughly 30 days on average. The number of all job ads which are considered at time T is the pseudo-stock of active job ads at time T. This approach has the advantage that it can account for gaps in scraping coverage as long as the missing ads are scraped at some point within 30 days of posting. One drawback of this approach is that for the first 30 days in the observed timeframe too few valid vacancies are available. In order to ease the comparison with JVS, we take the average of pseudo-stocks over the last month of each quarter and arrive at job ad stocks which are very roughly comparable with those of the JVS.

Some important variables are not available, such as the organization ID, visiting address of organization, title of ads and description of ads. There are other important variables for deriving variables, such as work unit and NACE. These variables are important because of the comparison between online job ads and JVS.

In some cases differing data patterns were identified when comparing CEDEFOP data with collected home OJA data due to the different approaches and methods in data pre-processing phases. (Annex BG).

More detailed metadata and processing information of the variables is necessary to facilitate further data analysis. The release of the raw text data will be very important for continuing the work. It is also important to know the proportion of data collected by API, crawled and scraped. The first urgent request on related open questions and requirements regarding the CEDEFOP data set has already been prepared by Germany. Answers have been provided by CEDEFOP (Annex DESTATIS).

Of equally great importance would be a massive expansion of processing capabilities of the Big Data Test Infrastructure. The current infrastructure is not capable of loading the full OJA-sample for the larger countries into memory. Any kind of Big Data analysis involving, for instance, machine learning or text analysis will require many times the memory and processing power that is currently available (Annex DESTATIS).

The detailed analysis reports by country can be found in the annex section of this document.

Potential statistical indicators

The official job vacancy statistics required by the EU regulation are typically not broken down by characteristics of the vacant job position, such as the geographical area/region where the vacant post is located and the profession required. However, these kinds of characteristics are considered as particularly useful by users. Additional information at higher frequencies than those of the JVS can be derived from the OJAs and gives useful insights into the dynamics within the quarter of the labour demand expressed by the online recruitment process.

Therefore, detailed *supplementary* indicators could be derived from OJAs. In particular,

Table 1: Potential indicators from OJAs

Indicator	**Frequency**	**Grouped by**
Number of available on-line job vacancies in a reference day	daily	economic activity, occupation, region, education
Number of available on-line job vacancies in a reference month	monthly	economic activity, occupation, region, education
Number of *newly* posted on-line job vacancies in a reference day	daily	economic activity, occupation, region, education
Number of *newly* posted on-line job vacancies in a reference	monthly	economic activity, occupation, region,

month		education
Average number of OJAs	weekly , monthly, quarterly	economic activity, occupation, region, education
Percentage changes over the same period of the previous year (year on year changes) of the number of OJAs	daily, weekly, monthly,	economic activity, occupation, region, education
Percentage changes over the same period of the previous quarter (quarter on quarter changes) of the number of OJAs	daily, weekly, monthly,	economic activity, occupation, region, education
Number of available on-line job vacancies at the JVs reference day	quarterly	economic activity, occupation
Average of daily pseudo-stocks (as approximation to the true stock) of job ads	daily, weekly, monthly (recommended), quarterly	industry sector, geo-region or required qualification
Indicator of labour market power by employers, Herfindahl-Hirschman Index for each quarter/commuting zone/occupation-cell (compare Azar et al. [3])	quarterly	commuting zone (functional urban area) & occupation

One area that shows some promise is to use the time series properties of OJA data to improve existing statistics [2]. The pilot has had modest success in predicting job vacancy survey values using OJA data, so these data could be used for producing flash estimates of labour demand. It may also be possible to use these time series properties to produce more frequent estimates, or even reduce the frequency of the survey. An important limiting factor of current CEDEFOP data is that this system holds data from 2018 onwards, so it will take several years at least to collect a reasonable time series.

CEDEFOP data investigated here clearly demonstrate a strong relationship between OJAs and socio-economic variables such as national and regional population sizes and GDPs. These relationships can be modelled with

promising outputs. Such models may potentially be used for predictive purposes. For instance, since OJAs are potentially instantly available, they might offer a data source for developing flash estimates of national and regional GDPs. The models and choice of variables should of course be studied in more detail and improved. The models also indicate that they offer the potential of predicting OJAs for countries without data for OJAs on the basis of their socio-economic variables. Such cases without CEDFOP OJAs data offer an interesting opportunity to test the models (Annex Swiss Federal Statistical Office).

The analysis also focused on the potential of OJA's contract type and working hours for estimating economic statistics:

- How do contract type and working hours explain Consumer Confidence (CCI) and Gross Domestic Product (GDP)?
- How does the overall count of job vacancies explain Consumer Confidence and Gross Domestic Product?

There does not seem to be a clear indication that contract type or working hours are a good predictor of the Consumer Confidence Index or GDP. However, there are data quality issues with the contract type variable. While there are promising results of Working Hours and GDP, there would need to be more work undertaken to the pre-processing of the CEDEFOP data to give more substantial conclusions. These results reinforce the need for unrestricted access to the text of the job ads and for an infrastructure capable of analysing such data. (Annex ONS).

Methodological framework

According to the plan of the WPB deliverables, the first version of the methodological framework for processing online job adverts data for official statistics was developed.

The purpose of this methodological framework is to describe at the conceptual level complete statistical online job adverts data (OJA) processing pipeline according to the four main phases: data ingestion, data pre-processing, information extraction and analysis, and producing statistical indicators. It will address key conceptual, methodological, technical models and organisational principles and guidelines in using OJA data in official statistics. The aim of this framework is therefore to:

- Provide a common language for various stakeholders
- Encourage adherence to common standards, specifications, methodologies
- Provide consistent methods for implementation
- Illustrate various implementation scenarios by identifying common components, processes and systems
- Provide a reference for statistical agencies and other parties to understand, discuss and compare OJA processing systems

Due to high complexity and heterogeneity of related OJA data pre-processing, this framework also discusses the role of a potential common OJA data collection and processing system (for the time being based on the OJA system developed by CEDEFOP) in the form of a future Web Intelligence Hub. The framework consists of two main methodological considerations around OJAs and ESS OJA reference production architecture. The latter is developed in close cooperation with WPF taking into account developments related to the Big Data Reference Architecture and Layers Framework (BREAL).

Prototype development and sharing

One of the objectives of this work package is also development and sharing of functional prototypes and methodologies related to collecting, processing and analysing OJA data. During the first half of the project WPB partners shared several prototypes which cover different phases of OJA processing pipeline: scraping, crawling, data pre-processing, text classification, deduplication, modelling and prediction, and data visualisation.

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Annex

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Annex: DESTATIS, Germany

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